

Histidine Catabolism to Glutamate in *Pseudomonas putida* KT2440 — Module/Pathway/Taxon Review

Target taxon: *Pseudomonas putida* KT2440 (organism code PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu00340 "Histidine metabolism" (resolved local bucket kegg:ppu00340; module area: amino acid metabolism) **Module reviewed:** Histidine catabolism to L-glutamate (generic reusable module; His → urocanate → imidazolone-propionate → N-formimidoyl-glutamate → L-glutamate)

1. Executive Summary

The histidine-catabolism-to-glutamate module is **fully satisfiable (COVERED)** in *Pseudomonas putida* KT2440. The organism uses the classical bacterial **five-step Hut "formate" route**, encoded in a single contiguous, single-copy gene cluster spanning **PP_5029–PP_5036**. The core catabolic reactions are: **HutH** (histidase, PP_5032, EC 4.3.1.3) → **HutU** (urocanase, PP_5033, EC 4.2.1.49) → **HutI** (imidazolonepropionase, PP_5030, EC 3.5.2.7) → **HutF** (formimidoylglutamate deiminase, PP_5036, EC 3.5.3.13) → **HutG** (N-formylglutamate deformylase, PP_5029, EC 3.5.1.68). The cluster additionally encodes the ancillary functions **HutT** (histidine transporter, PP_5031), **HutC** (pathway repressor, PP_5035), and **HutD** (PP_5034, Pfam PF05962). Every core enzyme is present in exactly one copy, and there are no genuine enzymatic gaps in the pathway.

The two lineage-specific terminal alternatives described by the generic module — the **Bacillus-type formimidoylglutamase route** (EC 3.5.3.8) and the **folate-coupled FTCD route** (glutamate formimidoyltransferase EC 2.1.2.5 + formimidoyltetrahydrofolate cyclodeaminase EC 4.3.1.4) — are **verified absent proteome-wide** and should be marked `not_expected_in_target_taxon`. KT2440 disposes of the formimino group exclusively via the HutF/HutG formate route, releasing ammonium and formate.

The most important curation issue is **scope contamination in the candidate gene list**. Of the 20 candidate genes, only 5 belong to this module (the *hut* catabolic genes). The remaining 15 are over-propagated from the shared KEGG map ppu00340, which bundles histidine **biosynthesis** and **degradation** in one bucket: the *his* biosynthesis genes (*hisG/D/C/B/H/A/F/I/E*, *hisZ*), the histidinol-phosphatases (*PP_3157*, *PP_5147*), a phosphoserine phosphatase (*PP_1721*), plus two unrelated cross-mapped genes — *gshA* (glutathione biosynthesis, *PP_0243*) and a tryptophan 2-monooxygenase (*PP_4983*, from *ppu00350*). The recommendation is to rebind the module to KEGG M00045 (histidine degradation, histidine → N-formiminoglutamate → glutamate) / MetaCyc PWY-5030 and mark the module `module_needs_revision` on boundary grounds only — the biology itself is complete and clean. Crucially, there is direct target-strain experimental evidence that the system is physiologically active: deletion of *hutT* (*PP_5031*) severely impairs growth of *KT2440* on histidine, and reconstituted *HutT* is a high-affinity histidine:H⁺ symporter (PMID: 34245008). Regulatory context is established for the species: histidine assimilation as a carbon/nitrogen source is under *CbrA/B* and *NtrC* control in *P. putida** (PMID: 20553554).

2. Target-Organism Pathway Definition

What the module includes

The module is the **catabolic degradation of L-histidine to L-glutamate**. In *KT2440* this is a linear five-reaction sequence with a defined terminal disposal of the formimino carbon:

1. **Histidine deamination** — histidine ammonia-lyase (histidase, *HutH*) releases NH₃ and produces *trans*-urocanate.
2. **Urocanate hydration** — urocanate hydratase (urocanase, *HutU*) produces 4-imidazolone-5-propanoate.
3. **Imidazolone ring opening** — imidazolonepropionase (*HutI*) hydrolyzes the ring to N-formimidoyl-L-glutamate (FIGLU).
4. **FIGLU deimination** — formimidoylglutamate deiminase (*HutF*) releases NH₃ and forms N-formyl-L-glutamate.
5. **Deformylation** — N-formylglutamate deformylase (*HutG*) releases formate and yields **L-glutamate**.

Net: L-histidine + 2 H₂O → L-glutamate + 2 NH₃ (as ammonium) + formate. The glutamate is fed into central nitrogen/carbon metabolism, which is why histidine serves as both a carbon and a nitrogen source in this organism.

Pathway boundaries — what to keep separate

- **Histidine biosynthesis (de novo, His pathway).** KEGG ppu00340 conflates biosynthesis and degradation. The *his* genes (*ATP-PRT*, *HisD*, *HisC*, *HisB*, *HisH/F/A*, *HisI/E*, *HisZ*) build histidine from PRPP and are not part of this catabolic module*. They must be scored separately.
- **One-carbon / folate metabolism.** In organisms using the FTCD route, the formimino group is transferred to tetrahydrofolate and enters C1 metabolism. **This linkage does not exist in KT2440** (FTCD absent); the formimino carbon leaves as free formate via HutG.
- **Glutathione biosynthesis.** *gshA* (glutamate–cysteine ligase) is cross-mapped only because glutamate is a shared metabolite; it is unrelated to histidine catabolism.
- **Tryptophan/indole metabolism.** PP_4983 (tryptophan 2-monooxygenase) belongs to ppu00350 and is a spurious cross-map.

Alternate names and database definitions

- **KEGG:** module M00045 "Histidine degradation, histidine => N-formiminoglutamate => glutamate" is the correct catabolic module (ppu00340 is the broader combined map).
- **MetaCyc:** PWY-5030 "L-histidine degradation III" (the HutF/HutG bacterial formate route).
- **Common names:** "hut pathway," "histidine utilization pathway," "histidine dissimilation." Genes: *hutH* (histidase), *hutU* (urocanase), *hutI* (imidazolonepropionase), *hutF* (formiminoglutamate/formimidoylglutamate deiminase), *hutG* (N-formylglutamate deformylase), *hutC* (repressor), *hutD*, *hutT* (transporter).

3. Expected Step Model and Coverage

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L-Histidine
| [1] HutH histidase (EC 4.3.1.3) PP_5032 COVERED (single copy)
▼
trans-Urocanate
| [2] HutU urocanase (EC 4.2.1.49) PP_5033 COVERED (single copy)
▼
4-Imidazolone-5-propanoate
| [3] HutI imidazolonepropionase (3.5.2.7) PP_5030 COVERED (single copy)
▼
N-Formimidoyl-L-glutamate (FIGLU)
| [4] HutF formimidoylglutamate deiminase PP_5036 COVERED (single copy)
| (EC 3.5.3.13) -- releases NH3
▼
N-Formyl-L-glutamate
| [5] HutG N-formylglutamate deformylase PP_5029 COVERED (single copy)
| (EC 3.5.1.68) -- releases formate
▼
L-Glutamate -> central metabolism

ANCILLARY (same operon, PP_5029–PP_5036):
HutT PP_5031 L-histidine:H+ symporter (transport) COVERED (direct KT2440 IDA)
HutC PP_5035 pathway repressor COVERED
HutD PP_5034 Pfam PF05962 (cupin/HutD domain) COVERED (accessory)

TERMINAL ALTERNATIVES (generic module) – NOT in KT2440:
Bacillus-type formimidoylglutamate (EC 3.5.3.8) not_expected_in_target_taxon
Folate FTCD: formimidoyltransferase (EC 2.1.2.5) not_expected_in_target_taxon
+ cyclodeaminase (EC 4.3.1.4) not_expected_in_target_taxon

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All five core catabolic steps plus transport and regulation are covered by single-copy candidate genes in one operon. No step is missing, ambiguous, or dependent on an undiscovered paralog. The module is **satisfiable end-to-end**.

4. Candidate Genes and Evidence

4.1 In-scope genes (the *hut* module) — high confidence

Gene	Locus	UniProt	Enzyme / role	EC	Module step	Copy no.	Assessment
hutH	PP_5032	Q88CZ7	Histidase (histidine ammonia-lyase)	4.3.1.3	1	1	COVERED
hutU	PP_5033	Q88CZ6	Urocanate hydratase (urocanase)	4.2.1.49	2	1	COVERED (alias cleanup needed)
hutI	PP_5030	Q88CZ9	Imidazolonepropionase	3.5.2.7	3	1	COVERED
hutF	PP_5036	Q88CZ3	Formimidoylglutamate deiminase	3.5.3.13	4	1	COVERED
hutG	PP_5029	Q88D00	N-formylglutamate deformylase	3.5.1.68	5	1	COVERED

Evidence type. Assignments rest on (a) UniProt/genome annotation of the contiguous PP_5029–PP_5036 cluster in the KT2440 proteome, and (b) strong congeneric biochemical genetics. The classic *Pseudomonas* work established exactly this five-step route — "successive formation of urocanate, imidazol-4-on-5-ylpropionate, N-formimino-l-glutamate, N-formyl-l-glutamate and glutamate" (PMID: 4146796) — and *P. fluorescens* SBW25 genetics showed that "inactivation of hutF eliminated the ability to grow on histidine, indicating that SBW25 degrades histidine by the five-step enzymatic pathway" (PMID: 17717196). Both are strong transfers to the congeneric KT2440, whose cluster carries the identical gene set. Genetic control of the *P. putida* histidine dissimilatory pathway was also mapped historically (PMID: 4405673).

Curation caveat on HutU (PP_5033). The metadata label carries a spurious secondary description ("Imidazolonepropionate hydrolase"). HutU is urocanate hydratase (EC 4.2.1.49); imidazolone-propionate hydrolysis is HutI's reaction. This alias should be cleaned to avoid confusing the EC 3.5.2.7 step with the EC 4.2.1.49 step.

4.2 Ancillary in-scope genes

Gene	Locus	UniProt	Role	Evidence	Assessment
hutT	PP_5031	—	L-histidine:H ⁺ symporter (major uptake)	Direct KT2440 IDA — <i>ΔhutT</i> growth defect; reconstituted transporter	COVERED (transport)
hutC	PP_5035	—	Hut pathway repressor (helix-turn-helix)	Homology to <i>P. putida</i> / <i>K. aerogenes</i> HutC	COVERED (regulation)
hutD	PP_5034	Q88CZ5	Accessory; Pfam PF05962 (HutD), RmlC-like cupin	Domain assignment (InterPro IPR010282)	COVERED (accessory)

HutT direct evidence. In KT2440, "deletion of hutT severely impairs growth of *P. putida* on histidine, suggesting that the encoded transporter is the major histidine uptake system of *P. putida*" (PMID: 34245008). This is the strongest, target-strain-specific functional confirmation that the *hut* system is active and that histidine is a genuine growth substrate.

HutD identification. PP_5034 was uncharacterized in the base label but sits in the 3' region of *hutC*. It carries Pfam PF05962 / InterPro IPR010282 (HutD domain) plus RmlC-like cupin folds, identifying it as **HutD**, the conserved accessory protein of the *hut* operon.

4.3 Out-of-scope candidates (histidine biosynthesis — over-propagated)

These are **de novo histidine biosynthesis** genes captured only because KEGG ppv00340 merges biosynthesis and degradation. They should be **excluded from this module's satisfiability scoring**.

Gene	Locus	Role (biosynthesis)	EC
hisG	PP_0965	ATP phosphoribosyltransferase	2.4.2.17
hisD	PP_0966	Histidinol dehydrogenase	1.1.1.23
hisC	PP_0967	Histidinol-phosphate aminotransferase	2.6.1.9
hisB	PP_0289	Imidazoleglycerol-phosphate dehydratase	4.2.1.19
hisH	PP_0290	IGP synthase glutaminase subunit	4.3.2.10 / 3.5.1.2
hisA	PP_0292	ProFAR isomerase	5.3.1.16
hisF	PP_0293	IGP synthase cyclase subunit	4.3.2.10
hisI	PP_5014	Phosphoribosyl-AMP cyclohydrolase	3.5.4.19
hisE	PP_5015	Phosphoribosyl-ATP pyrophosphatase	3.6.1.31
hisZ	PP_4890	ATP-PRT regulatory subunit	—
PP_3157	PP_3157	Histidinol-phosphatase	3.1.3.15
PP_5147	PP_5147	Histidinol-phosphatase	3.1.3.15
PP_1721	PP_1721	Phosphoserine phosphatase (broad 3.1.3.-)	3.1.3.-

4.4 Unrelated cross-mapped candidates

Gene	Locus	Actual role	Why mis-mapped
gshA	PP_0243	Glutamate–cysteine ligase (glutathione)	Shared metabolite (glutamate)
PP_4983	PP_4983	Tryptophan 2-monooxygenase	From ppu00350; spurious cross-map

5. Gaps, Ambiguities, and Likely Over-Annotations

No genuine enzymatic gaps

Every core catabolic step (1–5) has a dedicated single-copy gene in the PP_5029–PP_5036 operon. There are **no missing steps** and **no reliance on hypothetical paralogs**. The module does not require any `candidate_uncertain` enzymatic step.

Terminal alternatives confirmed absent (not gaps)

A proteome-wide UniProt search of KT2440 (organism_id 160488) returned **zero proteins** for: - EC 2.1.2.5 — glutamate formimidoyltransferase (FTCD, transfer half) - EC 4.3.1.4 — formimidoyltetrahydrofolate cyclodeaminase (FTCD, cyclodeaminase half) - EC 3.5.3.8 — formimidoylglutamase (Bacillus-type route) - Zero text hits for "formimino"/"formiminotransferase."

Therefore the folate-coupled and *Bacillus*-type routes are correctly `not_expected_in_target_taxon` — their absence is a **true negative**, not a coverage gap. KT2440 exclusively runs the HutF/HutG formate branch.

Single-copy status (contrast with a relative)

Each *hut* enzyme is present in exactly one copy in KT2440. This differs from *P. fluorescens* SBW25, which carries **two *hutH* copies** (one nonfunctional) ([PMID: 17717196](#)). KT2440 has no such paralog ambiguity, simplifying curation.

Over-annotation issues to flag

1. **KEGG map over-propagation (primary issue)**. 15 of 20 candidates are out-of-scope. The combined biosynthesis+degradation KEGG map is the root cause. This is a **module boundary problem**, not a biology problem.
 2. **HutU alias contamination** (see 4.1) — "imidazolonepropionate hydrolase" should be removed from PP_5033.
 3. **Broad-EC phosphatases** — PP_1721 (EC 3.1.3.-) and the two histidinol-phosphatases (PP_3157, PP_5147, EC 3.1.3.15) have broad/ambiguous mappings and belong to biosynthesis, not catabolism.
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6. Module and GO-Curation Recommendations

Step-level verdicts

Module step	Gene(s)	Verdict
1. Histidine deamination (HutH)	PP_5032	covered
2. Urocanate hydration (HutU)	PP_5033	covered
3. Imidazolone ring opening (HutI)	PP_5030	covered
4. FIGLU deimination (HutF)	PP_5036	covered
5. N-formylglutamate deformylation (HutG)	PP_5029	covered
Transport (HutT)	PP_5031	covered (direct KT2440 evidence)
Regulation (HutC)	PP_5035	covered
Accessory (HutD)	PP_5034	covered
Bacillus formimidoylglutamase route (EC 3.5.3.8)	—	not_expected_in_target_taxon
Folate FTCD route (EC 2.1.2.5 + 4.3.1.4)	—	not_expected_in_target_taxon

Module-level actions

- **Overall module verdict: COVERED.** The histidine→glutamate module is fully satisfiable via the single-copy HutF/HutG formate route.
- **module_needs_revision (boundaries only).** The module is currently bound to KEGG ppu00340, a combined biosynthesis+degradation map. **Rebind to KEGG M00045 / MetaCyc PWY-5030** and remove the 15 over-propagated candidates from module scoring (hisG/D/C/B/H/A/F/I/E, hisZ, PP_1721, PP_3157, PP_5147, gshA, PP_4983).
- **No new module documents needed** — the generic module structure already models the correct (formate) branch; KT2440 simply uses one of the three modeled alternatives.
- **No new GO terms required.** Existing GO terms cover all reactions: histidine catabolic process to glutamate, histidine ammonia-lyase activity (GO:0004397), urocanate hydratase activity (GO:0016153), imidazolonepropionase activity (GO:0050480), formimidoylglutamate deiminase activity (GO:0050415), N-formylglutamate deformylase activity (GO:0050129). The

hut genes should be checked to ensure these specific GO annotations are present rather than only broad parent terms.

7. Genes to Promote to Full [fetch-gene](#) Review

Gene	Locus	Reason for promotion
hutF	PP_5036	Terminal-branch determinant (EC 3.5.3.13); its presence is what fixes KT2440 in the formate route vs. alternatives — verify sequence/annotation directly
hutG	PP_5029	Terminal deformylase (EC 3.5.1.68); confirm activity annotation and that it is the true route terminus
hutD (PP_5034)	PP_5034	Newly identified as HutD from domain evidence (PF05962) but base label was "uncharacterized"; promote to formalize the annotation
hutU	PP_5033	Alias cleanup required ("imidazolonepropionate hydrolase" erroneous); confirm EC 4.2.1.49 only

Lower priority: HutH (PP_5032), HutI (PP_5030), and HutT (PP_5031) are well-supported (HutT has direct KT2440 evidence) and need review only if annotation harmonization is desired.

8. Mechanistic Interpretation (Synthesis)

KT2440 packages the entire histidine-utilization function into a single genomic neighborhood, a hallmark of the *Pseudomonas hut* system:

Genomic organization (PP_5029–PP_5036):

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PP_5029  PP_5030  PP_5031  PP_5032  PP_5033  PP_5034  PP_5035  PP_5036
hutG     hutI     hutT     hutH     hutU     hutD     hutC     hutF
(5.1.68) (3.5.2.7) (transp) (4.3.1.3) (4.2.1.49) (access) (repress) (3.5.3.13)
|         |         |         |         |         |         |
one contiguous, single-copy cluster; His as C+N source

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Functionally the flow is: **HutT imports histidine** → **HutH deaminates** it (NH₃ #1) → **HutU hydrates** urocanate → **HutI opens** the imidazolone ring to FIGLU → **HutF deiminates** FIGLU (NH₃ #2) to N-formyl-glutamate → **HutG deformylates** (formate) to **glutamate**. Glutamate then enters central metabolism, and its handling is itself regulated (AauR-AauS, [PMID: 17021207](#)). Expression of the pathway is repressed by HutC and integrated into global C/N control via CbrA/B and NtrC, which is why growth on histidine as a sole C/N source depends on this regulatory network ([PMID: 20553554](#), [25031426](#)). A notable maturation detail: urocanase (HutU) requires an α-ketobutyrate prosthetic group whose biosynthesis depends on threonine dehydratase ([PMID: 4154935](#)) — a dependency worth noting if urocanase activity is ever assayed but not relevant to gene-level module satisfiability.

The key curatorial insight is the mismatch between **biology** (a clean, complete, single-copy formate route) and **bookkeeping** (a KEGG bucket that dumps 15 unrelated biosynthesis/cross-mapped genes into the same list). The module is satisfiable; the candidate list is noisy.

9. Evidence Base

PMID	Organism	Relevance	Strength for KT2440
4146796	<i>Pseudomonas testosteroni</i>	Establishes the five-step formate route (urocanate → imidazolone-propionate → N-formimino-Glu → N-formyl-Glu → Glu)	Strong (route definition)
17717196	<i>P. fluorescens</i> SBW25	$\Delta hutF$ abolishes histidine growth → HutF-dependent five-step route; also documents duplicate <i>hutH</i>	Strong congeneric transfer
4405673	<i>P. putida</i>	Genetic control of the histidine dissimilatory pathway	Strong (same species)
34245008	<i>P. putida</i> KT2440	$\Delta hutT$ growth defect; HutT is the major histidine:H ⁺ symporter	Direct target strain
20553554	<i>P. putida</i>	CbrAB required for assimilation of histidine (and proline/arginine) as C/N sources	Direct species regulatory
25031426	<i>P. putida</i> / <i>P. aeruginosa</i>	CbrA/B/CrcZ(Y) carbon-catabolite-repression signaling operates similarly across species	Species-level context
2203754	<i>Klebsiella aerogenes</i>	HutC repressor sequence; notes strong similarity to <i>P. putida</i> HutC	Supports HutC (PP_5035) assignment
4154935	<i>P. putida</i>	Urocanase prosthetic group (α -ketobutyrate) biosynthesis via threonine dehydratase	Mechanistic detail on HutU maturation
9561727	<i>P. syringae</i> (+ <i>P. putida</i> assay)	<i>hut</i> operon (urocanase/histidase) upregulated at low temperature	Genus-level regulation
17021207	<i>P. putida</i> KT2440	AauR-AauS two-component control of acidic amino acid (Glu) uptake/metabolism	Downstream glutamate handling context

Direct vs. inferred summary. The strongest, *directly experimental* evidence for the target strain is the HutT transporter phenotype (PMID: 34245008) and the CbrAB regulatory requirement (PMID: 20553554), both confirming histidine is used as a C/N source in KT2440. The **enzymatic step assignments** rest on genome/UniProt annotation of the KT2440 cluster combined with strong biochemical genetics from congeneric *Pseudomonas* (PMID: 4146796, 17717196, 4405673) — high-confidence homology transfer, not direct KT2440 enzymology for each individual enzyme.

10. Limitations and Knowledge Gaps

- **Per-enzyme direct biochemistry in KT2440 is limited.** The step assignments for HutH/HutU/HutI/HutF/HutG rely on genome annotation plus congeneric genetics, not on purified-enzyme assays from KT2440 itself. Transfer is strong (same genus, identical cluster) but not experimentally verified enzyme-by-enzyme in the target strain.
 - **HutD function.** PP_5034 is confidently a HutD-domain protein (PF05962), but the precise biochemical role of HutD in the pathway remains generally under-characterized across bacteria.
 - **No KT2440-specific *hut*-regulation study** was found beyond the transporter and global CbrAB/NtrC work; fine regulatory detail (HutC operator sites, inducer specificity in KT2440) is inferred from other *Pseudomonas/Klebsiella*.
 - **Proteome search scope.** Absence of the FTCD/formimidoylglutamase routes was established by UniProt EC and text searches; this is robust but is an annotation-based negative rather than an experimental one.
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11. Proposed Follow-up Actions

1. **Rebind the module** from KEGG ppu00340 to KEGG M00045 / MetaCyc PWY-5030 and remove the 15 over-propagated candidates from satisfiability scoring. (Curation action — highest priority.)
2. **Clean the HutU (PP_5033) annotation** to remove the erroneous "imidazolonepropionate hydrolase" alias.

3. **Formalize HutD (PP_5034)** annotation from "uncharacterized" to "HutD (PF05962)" and promote to `fetch-gene`.
4. **Promote HutF, HutG, HutU, HutD** to full gene review (Section 7).
5. **Verify specific GO annotations** (GO:0004397, GO:0016153, GO:0050480, GO:0050415, GO:0050129) are attached to the respective *hut* genes rather than only broad parents.
6. **Optional experimental confirmation** (if a wet-lab arm exists): construct $\Delta hutH$, $\Delta hutF$, $\Delta hutG$ single mutants in KT2440 and confirm loss of growth on histidine as sole C/N source, mirroring the SBW25 $\Delta hutF$ result — this would upgrade the enzymatic steps from homology-inferred to direct target-strain evidence.

12. Key References

- [PMID: 34245008](#) — HutT is the major L-histidine transporter in *P. putida* KT2440 (direct target-strain evidence).
- [PMID: 17717196](#) — Genetic analysis of *hut* genes in *P. fluorescens* SBW25; $\Delta hutF$ abolishes histidine growth.
- [PMID: 4146796](#) — Degradation of L-histidine via the five-step formate route in *Pseudomonas*.
- [PMID: 4405673](#) — Genetic control of the histidine dissimilatory pathway in *P. putida*.
- [PMID: 20553554](#) — CbrB and amino-acid (histidine) assimilation in *P. putida*.
- [PMID: 25031426](#) — CbrA/B carbon-source management in *P. putida* / *P. aeruginosa*.
- [PMID: 2203754](#) — HutC repressor sequence; similarity to *P. putida* HutC.
- [PMID: 4154935](#) — Urocanase α -ketobutyrate prosthetic group biosynthesis in *P. putida*.
- [PMID: 17021207](#) — AauR-AauS control of glutamate metabolism in *P. putida* KT2440.
- [PMID: 9561727](#) — Temperature regulation of the *hut* operon in *Pseudomonas*.

Review complete. Overall module verdict: COVERED via the single-copy HutF/HutG formate route (PP_5029–PP_5036), with direct KT2440 evidence for transport and regulation, terminal alternatives verified absent, and candidate-list boundaries flagged for revision (rebind to KEGG M00045 / MetaCyc PWY-5030).

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